

SUSHANT LOKHANDE

San Jose, CA | +1 669-340-6655 | lokhandesushant014@gmail.com | sushantlokhande.me |
linkedin.com/in/sushantlokhande14 | github.com/sushantlokhande14

EXPERIENCE

San Jose State University | Graduate Research Assistant, Distributed Systems & Data Aug 2025 to May 2026

- Built an automated **telemetry pipeline** that collected, normalized, and analyzed experiment data end to end, cutting setup and analysis effort **55%**.
- Reduced rebuffering **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configs.
- Validated **14 interoperability scenarios** across networking, dependency, and **failure-recovery** paths to confirm recovery behavior before large experiment runs.
- Generated **136,000** training images from **17,000** binaries across **8** visual representations and benchmarked **6** model families on **1,700** held-out samples.

Turito | Software Engineering Intern, Backend Infrastructure Jun 2022 to Oct 2022

- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Slashed p95 **CPU 22%** by profiling high-frequency media paths and eliminating excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Reduced production diagnosis time **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.
- Lowered network egress **14%** by removing redundant media-path processing and simplifying routing and payload handling.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Cut investigation time about **40%** by building a pipeline that aggregated load-test output from Rust workers into **SQLite** with Python and served comparisons in Flask.
- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Raised equivalent-load signaling throughput about **31%** by surfacing concurrency bottlenecks that drove targeted backend fixes.

TECHNICAL SKILLS

Languages: Python, Java, C++17, Rust, SQL, TypeScript

Data Platforms: PostgreSQL, Redis, SQLite, MongoDB, Kafka, Hadoop, MapReduce, Hive, Qdrant, Neo4j

Distributed Systems: Temporal, gRPC, Tokio, asyncio, Docker, AWS, Linux, load testing

Tooling: pytest, CI/CD, Git, profiling, telemetry pipelines, FastAPI, Flask

PROJECTS

AutoGrade AI, Multi-Agent Evaluation Platform | *Python, Temporal, Kafka, gRPC, PostgreSQL, Qdrant*

- Orchestrated 4-dimension rubric grading (correctness, efficiency, resilience, quality) across multiple LLM agents on **Temporal** and **Kafka**, with confidence gates and human-review queues.

Melody Metrics, Big-Data Pipeline | *C++, Hadoop, MapReduce, Hive, Python, XGBoost*

- Processed **1M+** heterogeneous records through C++ **MapReduce** and **Hive** into a curated **140,000**-row dataset powering an **84%**-accurate 81-genre classifier.

Relay, LLM Inference Gateway | *Python, FastAPI, asyncio, SQLite, SSE*

- Cut p50 latency from **759 ms to 44 ms** over a **4,000**-request load test with an OpenAI-compatible gateway doing exact and **semantic caching**, failover, and rate limiting.

OPEN SOURCE

- **fission/fission**: 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration
- **NVIDIA/garak**: preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Database Systems, Distributed Computing, NoSQL, Advanced Algorithms, Computer Networks, Machine Learning

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)